

DET-001 | Multiple Failed Logins — Brute Force HIGH

ID: DET-001

MITRE ID: T1110.001

Data Source: SigninLogs

Severity ● HIGH

Tactic:Credential Access

Category IDENTITY

ADX MAPPING LOGIC:

SigninLogs → StormEvents mapping: State = 'User/Source', EventType = 'Login attempt type', DamageProperty = 'failure count proxy', bin(StartTime,5m) = time window. We simulate: each State is a 'user', each storm event is a 'login attempt', EventType with high count per State in a window = brute force pattern.

OBJECTIVE: Detect a single 'user' (State) generating more than 10 events within a 5-minute window — simulating brute force login attempts. In real Sentinel this runs on SigninLogs filtering ResultType != 0 (failed logins).

KQL QUERY (RUNNABLE ON ADX FREE TIER)

```
StormEvents
| where StartTime > ago(365d)
| summarize
FailedAttempts = count(),
DistinctSources = dcount(Source),
SourceList = make_set(Source, 5),
FirstSeen = min(StartTime),
LastSeen = max(StartTime)
by State, bin(StartTime, 5m)
| where FailedAttempts > 10
| extend TimeDeltaMins = datetime_diff('minute', LastSeen, FirstSeen)
| project
State, FailedAttempts, DistinctSources,
SourceList, FirstSeen, LastSeen, TimeDeltaMins
| order by FailedAttempts desc
| take 20
```

INVESTIGATION STEPS

1. In real env: check if any attempt succeeded (ResultType == 0).
2. Lookup source IP reputation in VirusTotal / AbuseIPDB.
3. Check if account belongs to a VIP or privileged user.
4. Review geolocation of source IPs for anomalies.

RESPONSE ACTIONS

- Enable MFA immediately if not already present.
- Block the source IP in Conditional Access.
- Reset the targeted user account password.
- Escalate if a successful login follows the failures.

Output for the above Query:

The screenshot shows the Microsoft Sentinel KQL query editor interface. At the top, there are buttons for 'Run', 'Recall', 'KQL tools', and 'help/Samples'. Below the buttons is a KQL query for 'StormEvents'. The query filters for events from the last 365 days, summarizes failed attempts by state and time bin, and projects specific fields. The results table below the query is empty, showing columns for State, FailedAttempts, DistinctSources, SourceList, FirstSeen, LastSeen, and TimeDeltaMins. The status bar at the bottom indicates the query was completed on 2026-06-01 at 13:02 UTC in 0.521 seconds.

```
1 StormEvents
2 | where StartTime > ago(365d)
3 | summarize
4 FailedAttempts = count(),
5 DistinctSources = dcount(Source),
6 SourceList = make_set(Source, 5),
7 FirstSeen = min(StartTime),
8 LastSeen = max(StartTime)
9 by State, bin(StartTime, 5m)
10 | where FailedAttempts > 10
11 | extend TimeDeltaMins = datetime_diff('minute', LastSeen, FirstSeen)
12 | project
13 State, FailedAttempts, DistinctSources,
14 SourceList, FirstSeen, LastSeen, TimeDeltaMins
15 | order by FailedAttempts desc
16 | take 20
```

State	FailedAttempts	DistinctSources	SourceList	FirstSeen	LastSeen	TimeDeltaMins
No Rows To Show						

KQL Detection Query in Sentinel:

SigninLogs

```
| where TimeGenerated > ago(1h)
| where ResultType != "0" // 0 = success
| summarize
FailedAttempts = count(),
DistinctIPs = dcount(IPAddress),
IPList = make_set(IPAddress),
FirstSeen = min(TimeGenerated),
LastSeen = max(TimeGenerated)
by UserPrincipalName, bin(TimeGenerated, 5m)
| where FailedAttempts > 10
| extend TimeDelta = LastSeen - FirstSeen
| project
UserPrincipalName, FailedAttempts,
DistinctIPs, IPList, FirstSeen, LastSeen, TimeDelta
| order by FailedAttempts desc
```